

Business Insights & Final Conclusions

ShopSphere – E-commerce Sales and Profitability Analysis

Company: ShopSphere

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1 Executive Summary of Key Insights

The 36 SQL queries executed against `shopsphere_sales` confirm management's original concern: **sales are healthy and growing, but profit is being eroded by a narrow set of structural issues** – excessive discounting, a handful of chronically unprofitable sub-categories and products, and weak performance concentrated in a few states. The six insights below summarise the findings developed in full in Section 8.

Insight 1 – Growth is real, but uneven

Total sales reached \$2,297,200.86 across 5,009 orders (793 unique customers), with sales climbing from \$484,247.50 in 2014 to \$733,215.26 in 2017. Growth was not linear: sales **dipped 2.83%** in 2015 before rebounding **29.47%** in 2016 and **20.36%** in 2017.

Insight 2 – Furniture sells well but barely profits

Furniture is the #2 category by sales (\$741,999.80) but converts that into only \$18,451.27 profit – a **2.49% margin**, far below Technology (17.40%) and Office Supplies (17.04%). The Tables sub-category alone loses **-\$17,725.48**.

Insight 3 – Discounting above 20% is reliably unprofitable

The correlation between discount and profit is -0.22 . Average profit margin falls from 34.02% at 0% discount to **-16.69%** at 21–40% discount and **-122.52%** at 60%+ discount – discounting is the single clearest lever on profitability in the data.

Insight 4 – Losses are concentrated in specific states

19 of 49 states operate at a net loss. Texas is the most striking case: it is the **3rd-largest state by sales** (\$170,188.05) yet the single biggest loss-maker (**-\$25,729.36**), followed by Ohio, Pennsylvania and Illinois.

Insight 5 – Customer retention is already strong

98.5% of customers placed more than one order (6.32 orders/customer on average). The Consumer segment drives the most profit (\$134,119.21) but Home Office customers are the most profitable per dollar of sales (14.03% margin vs. 11.55% for Consumer).

Insight 6 – Shipping mode is a volume story, not a profitability problem

Every shipping mode is profitable. Standard Class generates the most profit (\$164,088.79, 57% of total) because it carries the most volume, even though it is the slowest option (5.01 average days vs. 0.04 for Same Day).

2 Analytical Findings

2.1 Overall Performance – KPI Summary

KPI	Value
Total Sales	\$2,297,200.86
Total Profit	\$286,397.02
Overall Profit Margin	12.47%
Total Orders (distinct)	5,009
Average Order Value	\$458.61
Total Unique Customers	793
Average Discount	15.62%
Sales Growth, 2016 → 2017	+20.36%

2.2 Which categories generate the highest sales and profit?

Sales are fairly evenly split across the three categories, but profit is not:

Category	Sales	Profit	Margin
Technology	\$836,154.03	\$145,454.95	17.40%
Office Supplies	\$719,047.03	\$122,490.80	17.04%
Furniture	\$741,999.80	\$18,451.27	2.49%

Furniture ranks #2 by revenue but last by a wide margin on profitability. Its average discount (17.39%) is also the highest of the three categories (Office Supplies 15.73%, Technology 13.23%), directly linking Insight 2 to Insight 3.

2.3 Which products consistently lose money?

At the sub-category level, three of seventeen sub-categories are unprofitable overall: Tables (-\$17,725.48), Bookcases (-\$3,472.56) and Supplies (-\$1,189.10) – while Copiers (\$55,617.82), Phones (\$44,515.73) and Accessories (\$41,936.64) are the strongest profit contributors.

At the individual product level, the five biggest loss-making SKUs are:

Product	Total Profit
Cubify CubeX 3D Printer, Double Head Print	-\$8,879.97
Lexmark MX611dhe Monochrome Laser Printer	-\$4,589.97
Cubify CubeX 3D Printer, Triple Head Print	-\$3,839.99
Chromcraft Bull-Nose Wood Oval Conf. Tables & Bases	-\$2,876.12
Bush Advantage Collection Racetrack Conf. Table	-\$1,934.40

Two Cubify 3D-printer variants together lose over \$12,700, and four of the remaining five worst products are conference/meeting tables – confirming that Tables is a *structural* problem, not a couple of isolated outliers.

2.4 Regional and state performance

Region	Sales	Profit	Margin
West	\$725,457.82	\$108,418.45	14.95%
East	\$678,781.24	\$91,522.78	13.49%
South	\$391,721.91	\$46,749.43	11.93%
Central	\$501,239.89	\$39,706.36	7.92%

Nineteen states run at a net loss. The ten worst are:

State	Total Profit
Texas	-\$25,729.36
Ohio	-\$16,971.38
Pennsylvania	-\$15,559.96
Illinois	-\$12,607.89
North Carolina	-\$7,490.91
Colorado	-\$6,527.86
Tennessee	-\$5,341.69
Arizona	-\$3,427.92
Florida	-\$3,399.30
Oregon	-\$1,190.47

California (\$76,381.39), New York (\$74,038.55) and Washington (\$33,402.65) are the top three profit-generating states. Texas stands out as the priority case: it is a top-3 state by *sales* yet the single largest loss-maker, pointing to a state-level discounting or cost issue rather than weak demand.

2.5 Does discounting reduce profitability?

Yes – clearly and measurably. The overall discount-to-profit correlation is -0.22 , and profit margin degrades sharply once discounts exceed 20%:

Discount Level	Avg. Margin	Total Profit	Line Items
0% (No Discount)	34.02%	\$320,987.60	4,798
1–20%	17.44%	\$100,785.47	3,803
21–40%	-16.69%	-\$35,817.47	460
41–60%	-63.41%	-\$28,944.19	215
60%+	-122.52%	-\$70,614.40	718

Every discount band above 20% is, on average, unprofitable. Discounted line items above 20% number 1,393 – roughly 14% of all transactions – yet they erase \$135,375.86 of profit that the undiscounted and lightly-discounted business would otherwise retain.

2.6 Who are the most valuable customers?

793 unique customers placed 5,009 distinct orders (6.32 orders/customer on average), and 98.5% of customers ordered more than once, indicating strong underlying loyalty.

Customer ID	Total Profit
TC-20980	\$8,981.32
RB-19360	\$6,976.10
SC-20095	\$5,757.41
HL-15040	\$5,622.43
AB-10105	\$5,444.81

By segment, the Consumer group is the largest and most profitable in absolute terms:

Segment	Customers	Profit	Margin
Consumer	409	\$134,119.21	11.55%
Corporate	236	\$91,979.13	13.03%
Home Office	148	\$60,298.68	14.03%

Note the top sales customer (SM-20320, \$25,043.05) does not appear among the top five by profit – a sign of high revenue built on lower-margin, likely higher-discount purchases, consistent with Insight 3.

2.7 Which shipping mode performs best?

Ship Mode	Avg. Days	Total Profit
Standard Class	5.01	\$164,088.79
Second Class	3.24	\$57,446.64
First Class	2.18	\$48,969.84
Same Day	0.04	\$15,891.76

All four modes are profitable; Standard Class leads purely on volume (it carries the bulk of orders), not on a higher per-order margin. There is no evidence that a particular shipping mode is a source of the profit decline.

2.8 Which months and quarters generate the highest revenue?

The best single month was **November 2017** (\$118,447.83 sales); the best single profit month was **December 2016** (\$17,885.31). A consistent Q4 seasonal peak is visible in every year of the dataset:

Year	Annual Sales	YoY Growth	Q4 Sales
2014	\$484,247.50	–	\$179,627.73
2015	\$470,532.51	-2.83%	\$182,297.01
2016	\$609,205.60	+29.47%	\$236,098.75
2017	\$733,215.26	+20.36%	\$280,054.07

Q4 is the strongest quarter in all four years and has grown every year – a reliable, repeatable seasonal pattern that inventory and marketing planning can be built around.

3 Business Recommendations

Optimize Discounts & Improve Pricing Strategy

Cap standard discount approval at 20%. Anything beyond that threshold should require explicit managerial sign-off, since every band above 20% runs at a negative average margin (Section 8.5). This single change targets \$135,375.86 in erased profit.

Address Loss-Making Products

Re-price or discontinue the Tables sub-category (-\$17,725.48) and review the five specific SKUs identified in Section 8.3, especially the two Cubify 3D-printer variants and the conference-table line, which account for a disproportionate share of total losses.

Improve Regional Performance

Investigate Texas, Ohio, Pennsylvania and Illinois specifically – all combine meaningful sales volume with outsized losses, suggesting a regional discounting or fulfillment-cost issue rather than weak demand. Use the West region's discipline (14.95% margin, the highest of any region) as the internal benchmark.

Increase Overall Profit

Rebalance category-level promotional spend away from Furniture, which already carries the highest average discount (17.39%) and the lowest margin (2.49%) of the three categories, toward Technology and Office Supplies, where discounting is lighter and margins are healthiest.

Increase Customer Retention

Retention is already a strength (98.5% repeat-purchase rate). Formalize it: build a loyalty or account-management program around the top profit customers identified in Section 8.6 (TC-20980, RB-19360, SC-20095, HL-15040, AB-10105) and the Home Office segment, which converts sales to profit most efficiently (14.03% margin).

Plan Around Seasonality

Use the consistent Q4 revenue peak (Section 8.8) to time inventory build-up, staffing and marketing spend, since the pattern has held, and grown, in every year of the dataset.

4 Conclusion

The data confirms management's original hypothesis: **ShopSphere's profit challenge is not a demand problem – sales grew in three of the last four years, including 20–30% growth in each of the last two years – it is a margin discipline problem.** Profitability is being narrowed from three directions that all point back to controllable business decisions rather than market conditions:

1. **Discount policy:** any discount above 20% is, on average, unprofitable, and 14% of all line items fall into that range.
2. **Product mix:** a small, identifiable set of sub-categories and SKUs (Tables, Bookcases, Supplies, and specific printer/table products) consistently lose money and can be individually re-priced, renegotiated, or discontinued.
3. **Regional execution:** losses are concentrated in a handful of states (led by Texas, Ohio, Pennsylvania and Illinois) rather than being spread evenly across the country, which makes the problem addressable at a regional level.

On the positive side, the business is not starting from a weak foundation: customer retention is strong (98.5% repeat-purchase rate), Technology and Office Supplies are both healthy and growing profit centers, every shipping mode is profitable, and demand follows a reliable, plannable seasonal pattern. The recommended actions in Section 9 are therefore targeted and tactical – discount governance, product-level pricing fixes, and regional review – rather than a broad restructuring, and are expected to materially narrow the gap between ShopSphere’s sales growth and its profit growth going forward.